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Editors
npj Quantum Information

Re: Article submission to the Collection "Quantum machine learning: understanding capabilities, limitations, and perspectives for quantum advantage"

Dear Editors,

We submit "Conditional validity of quantum event classifiers under collider systematics and quantum estimation uncertainty" for consideration as an Article in the Collection named above. Rather than presupposing quantum advantage, the manuscript asks a prior question for deployable QML: under what observable information can a performance or scientific-inference claim be supported when both the scientific environment and the quantum estimator are uncertain?

The work makes a connection that is not captured by nominal QML benchmarking. Collider systematics determine whether the scientific claim is identifiable; finite-shot and noisy quantum-kernel evaluation additionally randomizes the implemented Gram matrix, refit, calibration, and operating point. We integrate these layers in one fail-closed framework and distinguish claims about the realized deployment from claims anchored to an ideal quantum kernel. This allows the paper to say exactly what quantum measurement changes without attributing ordinary stochasticity exclusively to quantum systems.

The results directly address the Collection's emphasis on capabilities and limitations. Matched classical controls remove every apparent quantum-specific performance and sensing effect, so we claim no quantum advantage. What remains quantum-specific is an experimentally quantified, measurement-induced deployment uncertainty whose effects propagate through the full decision pipeline. The collider setting is essential as a stringent scientific deployment environment: it supplies physics weights, invisible yield shifts, control regions, finite templates, and downstream signal-strength inference. Evidence spans disjoint simulated worlds, CMS open data, and a deliberately micro-scale IBM QPU demonstration, with all scope limits stated explicitly.

The manuscript combines formal identifiability and anytime-valid per-claim guarantees with registered falsifiers, immutable manifests, public code and data artifacts, and a complete correction trail. Several registered falsifiers fired; each adverse result was retained and used to narrow the claims. We believe this combination of rigorous negative controls, quantum deployment semantics, and physics-level validation will be useful to the journal's interdisciplinary QML readership.

For transparency, the same author group has released a distinct manuscript, "Sharp Target-Domain Certificates for Quantum-Kernel Advantage under Distribution Shift" (Zenodo DOI 10.5281/zenodo.21776862). That work studies finite-target partial identification of predictive advantage against a prespecified classical-kernel family on unrelated cybersecurity and tabular datasets. It shares no datasets, experiments, theorems, estimands, codebase, or reported results with the present collider-inference study. We provide it as related material so the Editors can assess the distinction directly.

This manuscript is not under consideration elsewhere. All authors have approved the manuscript and its submission. The authors declare no financial or non-financial competing interests.

Sincerely,

Roberto Fernández-Barrios
Corresponding author, on behalf of all authors